

Chapter 28: Tables and Figures

Table 28.3

Adjustment to Free-Flow Speed for Lane Width on a Freeway

Lane Width (ft)	Reduction in Free-Flow Speed, f_{LW} (mi/h)
≥ 12	0.0
11	1.9
10	6.6

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Table 28.4

Adjustment to Free-Flow Speed for Lateral Clearance on a Freeway

Right Shoulder Lateral Clearance (ft)	Reduction in Free-Flow Speed, f_{LC} (mi/h)			
	<i>Lanes in One Direction</i>			
	2	3	4	≥ 5
≥ 6	0.0	0.0	0.0	0.0
5	0.6	0.4	0.2	0.1
4	1.2	0.8	0.4	0.2
3	1.8	1.2	0.6	0.3
2	2.4	1.6	0.8	0.4
1	2.0	2.0	1.0	0.5
0	3.6	2.4	1.2	0.6

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Table 28.5

Adjustment to Free-Flow Speed for Total Lateral Clearance on a Multilane Highway

4-Lane Multilane Highways		6-Lane Multilane Highways	
Total Lateral Clearance (ft)	Reduction in Free-Flow Speed, f_{LC} (mi/h)	Total Lateral Clearance (ft)	Reduction in Free-Flow Speed, f_{LC} (mi/h)
≥ 12	0.0	≥ 12	0.0
10	0.4	10	0.4
8	0.9	8	0.9
6	1.3	6	1.3
4	1.8	4	1.7
2	3.6	2	2.8
0	5.4	0	3.9

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Table 28.6

Adjustment to Free-Flow Speed for Median Type on Multilane Highways

Median Type	Reduction in Free-Flow Speed, f_M (mi/h)
Undivided	1.6
TWLTL	0.0
Divided	0.0

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Table 28.7

Adjustment to Free-Flow Speed for Access-Point Density on a Multilane Highway

Access Density (Access Points/Mi)	Reduction in Free-Flow Speed, f_A (mi/h)
0	0.0
10	2.5
20	5.0
30	7.5
≥ 40	10.0

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Table 28.9Factors Considered in *CAF* and *SAF*

Condition	CAF	SAF
Weather	Yes	Yes
Incidents	Yes	No
Driver Population	Yes	Yes
Work Zones	Yes	Yes

Table 28.10Capacity Adjustment Factors (*CAF*) for Inclement Weather

Weather	Definition of Conditions	CAF for Unadjusted FFS of: (mi/h)				
		55	60	65	70	75
Medium Rain	>0.10–0.25 in/h	0.94	0.93	0.92	0.91	0.90
Heavy Rain	>0.25 in/h	0.89	0.88	0.86	0.84	0.82
Light Snow	>0.00–0.05 in/h	0.97	0.96	0.96	0.95	0.95
Light-Medium Snow	0.05–0.10 in/h	0.95	0.94	0.92	0.90	0.88
Medium-Heavy Snow	>0.10–0.50 in/h	0.93	0.91	0.90	0.88	0.87
Heavy Snow	>0.50 in/h	0.80	0.78	0.76	0.74	0.72
Severe Cold	<–4°F	0.93	0.92	0.92	0.91	0.90
Low Visibility	0.50–0.99 mi	0.90	0.90	0.90	0.90	0.90
Very Low Visibility	0.25–0.49 mi	0.88	0.88	0.88	0.88	0.88
Minimal Visibility	<0.25 mi	0.90	0.90	0.90	0.90	0.90
Normal Weather	None of the Above	1.00	1.00	1.00	1.00	1.00

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Table 28.11Speed Adjustment Factors (*SAF*) for Inclement Weather

Weather	Definition of Conditions	SAF for Unadjusted FFS of: (mi/h)				
		55	60	65	70	75
Medium Rain	>0.10 – 0.25 in/h	0.96	0.95	0.94	0.93	0.93
Heavy Rain	>0.25 in/h	0.94	0.93	0.93	0.92	0.91
Light Snow	>0.00 – 0.05 in/h	0.94	0.92	0.89	0.87	0.84
Light-Medium Snow	0.05 – 0.10 in/h	0.92	0.90	0.88	0.86	0.83
Medium-Heavy Snow	>0.10 – 0.50 in/h	0.90	0.88	0.86	0.84	0.82
Heavy Snow	>0.50 in/h	0.88	0.86	0.85	0.83	0.81
Severe Cold	< – 4°F	0.95	0.95	0.94	0.94	0.92
Low Visibility	0.50 – 0.99 mi	0.96	0.95	0.94	0.94	0.93
Very Low Visibility	0.25 – 0.49 mi	0.95	0.94	0.93	0.92	0.91
Minimal Visibility	<0.25 mi	0.95	0.94	0.93	0.92	0.91
Normal Weather	None of the Above	1.00	1.00	1.00	1.00	1.00

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Table 28.12*CAF* for the Effect of Traffic Incidents

Number of Lanes	No Incidents	<i>CAF</i> for Incidents with the Following Effects				
		Shoulder Closed	1 Lane Closed	2 Lanes Closed	3 Lanes Closed	4 Lanes Closed
2	1.00	0.81	0.70	NA	NA	NA
3	1.00	0.83	0.74	0.51	NA	NA
4	1.00	0.85	0.77	0.50	0.52	NA
5	1.00	0.87	0.81	0.67	0.50	0.50
6	1.00	0.89	0.85	0.75	0.52	0.52
7	1.00	0.89	0.88	0.80	0.63	0.63
8	1.00	0.89	0.89	0.84	0.66	0.66

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Table 28.13*CAF* and *SAF* for Non-Standard Driver Populations

Driver Familiarity Classification	<i>CAF</i>	<i>SAF</i>
All Familiar Drivers	1.0000	1.000
Mostly Familiar Drivers	0.968	0.975
Balanced Mix of Familiar and Unfamiliar Drivers	0.939	0.950
Mostly Unfamiliar Drivers	0.898	0.913
All Unfamiliar Drivers	0.852	0.863

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Table 28.14

Values of the Lane Closure Severity Index

Lanes in Normal Operation (N)	Open Lanes Through Work Zone (N_o)	Open Ratio (OR)	<i>LCSI</i>
3	3	1.00	0.33
2	2	1.00	0.50
4	3	0.75	0.44
3	2	0.67	0.75
4	2	0.50	1.00
2	1	0.50	2.00
3	1	0.33	3.00
4	1	0.25	4.00

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Table 28.15Maximum Service Flow Rates (*MSF*) for Freeways (pc/h/ln)

<i>FFS</i> (mi/h)	Level of Service				
	A	B	C	D	E
75	820	1,310	1,750	2,110	2,400
70	770	1,250	1,690	2,080	2,400
65	710	1,170	1,630	2,030	2,350
60	660	1,080	1,560	2,010	2,300
55	600	990	1,430	1,900	2,250

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Table 28.16Maximum Service Flow Rates (*MSF*) for Multilane Highways (pc/h/ln)

<i>FFS</i> (mi/h)	Level of Service				
	A	B	C	D	E
70	760	1,260	1,700	2,020	2,300
65	710	1,170	1,630	2,000	2,250
60	660	1,080	1,550	1,980	2,200
55	600	990	1,430	1,850	2,100
50	550	900	1,300	1,710	2,000
45	290	810	1,170	1,550	1,900

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Table 28.17

Passenger Car Equivalents for Heavy Vehicles in General Terrain Segments

Passenger Car Equivalent	Type of Terrain	
	Level	Rolling
E_{HV}	2.0	3.0

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Table 28.18

Passenger Car Equivalents for a Typical Mix of Heavy Vehicles on Specific Grades

% Grade	Length (mi)	Percentage of Trucks (Including Buses & RVs)							
		2%	4%	5%	6%	8%	10%	15%	≥20%
≤0	ALL	2.39	2.18	2.12	2.07	2.01	1.96	1.89	1.85
2	0.125	2.67	2.32	2.23	2.17	2.08	2.03	1.95	1.89
	0.375	3.63	2.82	2.64	2.52	2.35	2.25	2.10	2.02
	0.625	4.12	3.08	2.85	2.69	2.49	2.36	2.18	2.08
	0.875	4.37	3.21	2.96	2.78	2.56	2.42	2.22	2.11
	1.250	4.53	3.29	3.02	2.84	2.60	2.45	2.24	2.13
	≥1.500	4.58	3.31	3.04	2.86	2.61	2.46	2.46	2.14
2.5	0.125	2.75	2.36	2.27	2.20	2.11	2.04	1.95	1.90
	0.375	4.01	3.02	2.80	2.65	2.46	2.33	2.15	2.06
	0.625	4.66	3.35	3.08	2.88	2.64	2.48	2.36	2.15
	0.875	4.99	3.52	3.21	3.00	2.73	2.56	2.32	2.19
	1.250	5.20	3.64	3.30	3.08	2.79	2.60	2.35	2.22
	≥1.500	5.26	3.67	3.33	3.10	2.80	2.62	2.36	2.23
3.5	0.125	2.93	2.45	2.34	2.26	2.16	2.09	1.98	1.92
	0.375	4.86	3.46	3.16	2.96	2.69	2.53	2.30	2.18
	0.625	5.88	3.99	3.59	3.32	2.98	2.76	2.46	2.31
	0.875	6.40	4.75	3.81	3.51	3.12	2.88	2.55	2.38
	1.250	6.74	5.15	3.96	3.63	3.21	2.96	2.60	2.42
	≥1.500	6.83	5.27	3.99	3.66	3.24	2.98	2.62	2.44
4.5	0.125	3.13	2.56	2.43	2.21	2.13	2.01	2.01	1.95
	0.375	5.88	3.99	3.59	2.98	2.76	2.46	2.46	2.31
	0.625	7.35	4.75	4.22	3.39	3.10	2.71	2.71	2.51
	0.875	8.11	5.15	4.54	3.60	3.27	2.83	2.83	2.61
	≥1.000	8.33	5.27	4.63	3.66	3.33	2.87	2.87	2.64
5.5	0.125	3.27	2.69	2.53	2.42	2.28	2.19	2.95	1.98
	0.375	7.09	4.62	4.11	3.76	3.31	3.04	2.66	2.47
	0.625	9.13	5.68	4.97	4.49	3.88	3.51	3.00	2.74
	0.875	10.21	6.24	5.43	4.88	4.18	3.76	3.18	2.89
	≥1.000	10.52	6.41	5.57	5.00	4.27	3.83	3.24	2.93
6	0.125	3.51	2.76	2.59	2.47	2.32	2.22	2.08	2.00
	0.375	7.78	4.98	4.40	4.01	3.51	3.20	2.78	2.56
	0.625	10.17	6.23	5.42	4.87	4.17	3.75	3.18	2.88
	0.875	11.43	6.88	5.95	5.32	4.53	4.04	3.39	3.06
	≥1.000	11.81	7.08	6.11	5.46	4.64	4.13	3.45	3.11

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Table 28.19

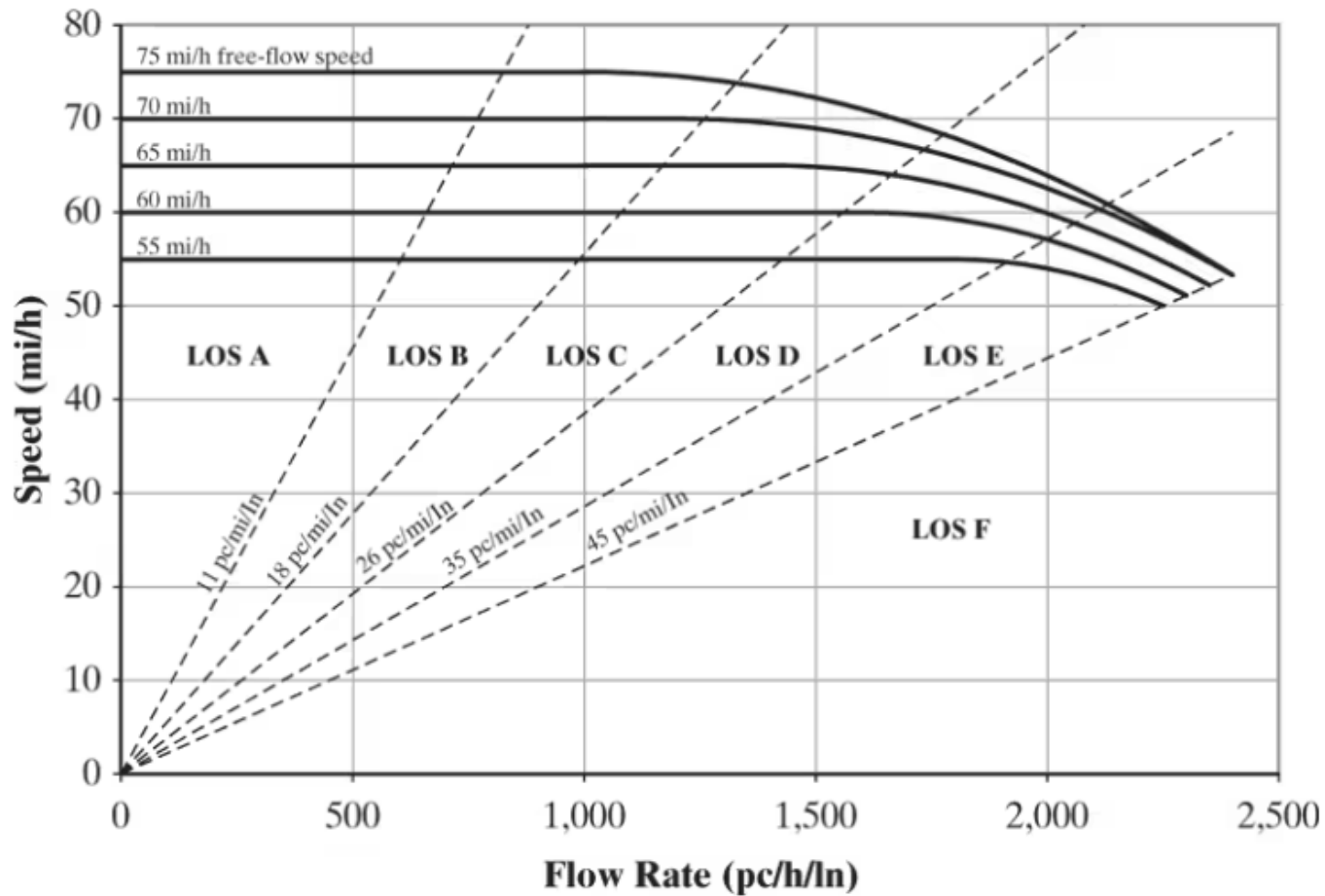
Passenger Car Equivalents for Atypical Mixes of Heavy Vehicles on Specific Grades

% Grade	Length (mi)	30% SUT – 70% TT (Heavy Mix)				50% SUT – 50% TT (Equal Mix)			
		Percentage of Trucks (Including Buses & RVs)							
		2%	5%	10%	15%	2%	5%	10%	15%
≤0	ALL	2.62	2.30	2.12	2.04	2.67	2.31	2.11	2.02
2	0.125	2.62	2.30	2.12	2.04	2.67	2.31	2.11	2.02
	0.375	3.76	2.78	2.38	2.22	3.76	2.77	2.36	2.20
	0.625	4.47	3.08	2.54	2.34	4.32	3.01	2.49	2.29
	0.875	4.80	3.22	2.61	2.39	4.57	3.11	2.55	2.33
	1.250	5.00	3.30	2.66	2.42	4.71	3.17	2.58	2.36
	≥1.500	5.04	3.32	2.67	2.43	4.74	3.19	2.59	2.36
2.5	0.125	2.62	2.30	2.12	2.04	2.67	2.31	2.11	2.02
	0.375	4.11	2.93	2.46	2.28	4.10	2.92	2.44	2.26
	0.625	5.04	3.32	2.67	2.43	4.84	3.23	2.61	2.38
	0.875	5.48	3.51	2.77	2.50	5.17	3.37	2.69	2.43
	1.250	5.73	3.61	2.83	2.54	5.36	3.45	2.73	2.47
	≥1.500	5.80	3.64	2.84	2.55	5.40	3.47	2.74	2.47
3.5	0.125	2.62	2.30	2.12	2.04	2.67	2.31	2.11	2.02
	0.375	4.88	2.93	2.63	2.41	4.89	3.25	2.62	2.39
	0.625	6.34	3.32	2.97	2.64	6.05	3.75	2.89	2.58
	0.875	7.03	3.51	3.12	2.76	6.58	3.97	3.01	2.67
	1.250	7.44	3.61	3.22	2.82	6.88	4.10	3.09	2.72
	≥1.500	7.53	3.64	3.24	2.84	6.95	4.13	3.10	2.73
4.5	0.125	2.62	2.30	2.12	2.04	2.67	2.31	2.11	2.02
	0.375	5.80	3.64	2.84	2.55	5.83	3.65	2.84	2.55
	0.625	7.90	4.53	3.32	2.90	7.53	4.38	3.24	2.83
	0.875	8.91	4.96	3.56	3.07	8.32	4.72	3.42	2.97
	≥1.000	9.19	5.08	3.62	3.11	8.53	4.81	3.47	3.00
	5.5	0.125	2.62	2.30	2.12	2.04	2.67	2.31	2.11
0.375		6.87	4.10	3.09	2.73	6.97	4.14	3.11	2.74
0.625		9.78	5.33	3.76	3.21	9.37	5.16	3.67	3.14
0.875		11.20	5.94	4.09	3.45	10.49	5.65	3.93	3.34
≥1.000		11.60	6.11	4.18	3.51	10.80	5.78	4.01	3.39
6		0.125	2.62	2.30	2.12	2.04	2.67	2.31	2.11
	0.375	7.48	4.36	3.23	2.73	7.64	4.43	3.26	2.85
	0.625	10.87	5.79	4.01	3.39	10.45	5.63	3.92	3.33
	0.875	12.54	6.51	4.40	3.67	11.78	6.20	4.24	3.56
	≥1.000	13.02	6.71	4.51	3.75	12.15	6.36	4.85	3.62

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Figure 28.6

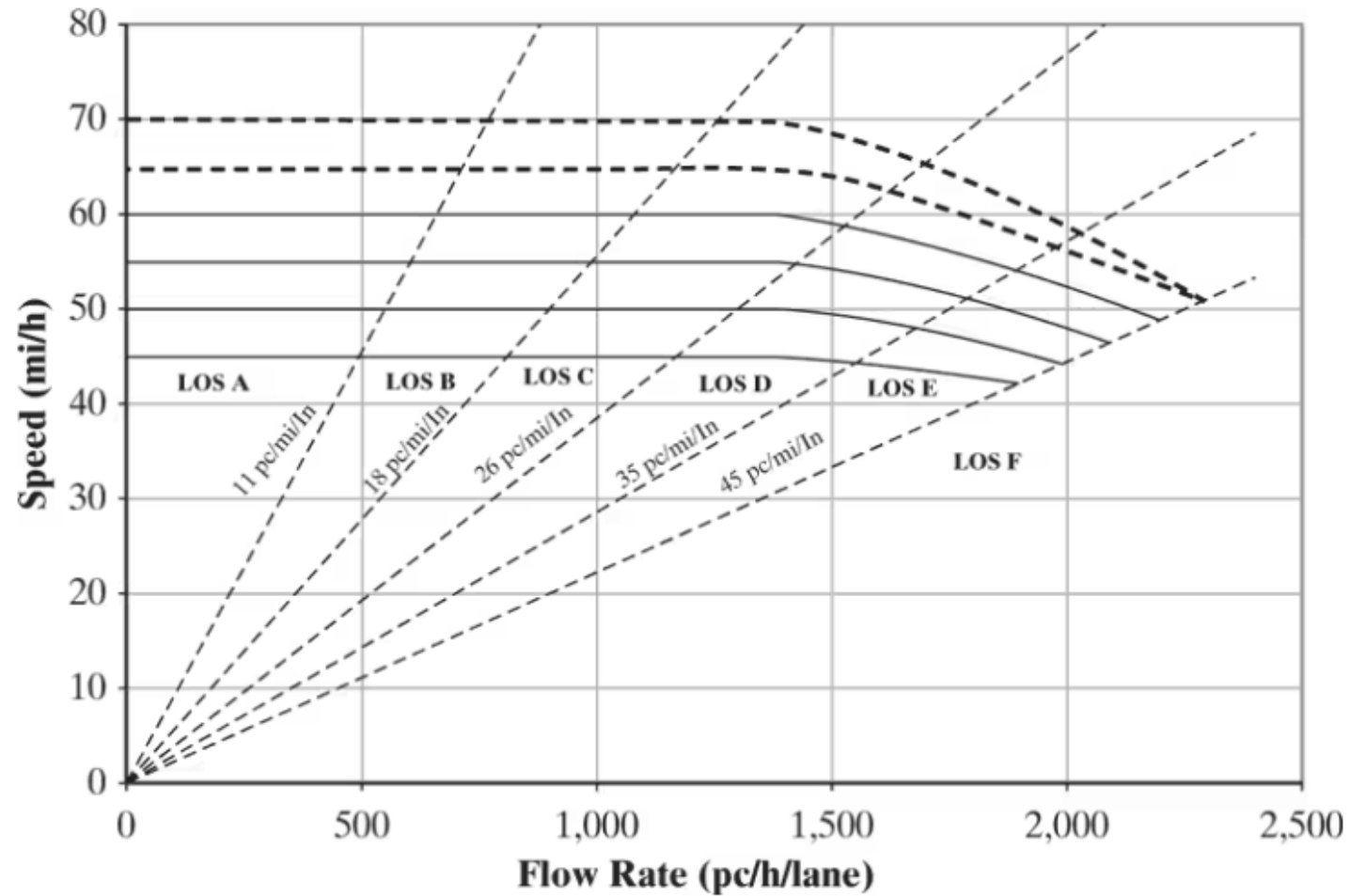
Example Base Speed-Flow Curves for Freeways



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Figure 28.7

Example Base Speed-Flow Curves for Multilane Highways



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